ETHAN HUECKER

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RESEARCH INTERESTS

Unconventional superconductivity, quantum geometry, and effective field theories of condensed matter in general.

EDUCATION

University of Florida 2023 - Present
PhD Physics

University of Florida 2023 - 2025
MS Physics - GPA 4.00/4.00

University of Cincinnati 2019 - 2023
BS Physics - GPA 3.85/4.00

PUBLICATIONS

- **E. Huecker**, M. Ye, Y. Wang, *Orbital magnetic susceptibility and de Haas-van Alphen effect of a flat band from quantum geometry*, arxiv:2610.12193 (2026).
- **E. Huecker**, Y. Wang, *Vestigial d-wave charge-4e Superconductivity from Bidirectional Pair Density Waves*, Phys. Rev. B **113**, 045121 (2026).
- **E. Huecker**, Y. Komijani, *Spin fractionalization in a Kondo-lattice superconductor heterostructure*, Phys. Rev. B **108**, 195120 (2023).

TEACHING EXPERIENCE

Lecturer, University of Florida

- *PHY 2049, a second semester calculus-based physics course.* Summer 2026

Teaching Assistant, University of Florida

- *PHY 2048, a first semester calculus-based physics course.* Fall 2026
- *PHY 2048, a first semester calculus-based physics course.* Spring 2026
- *PHY 2053, a first semester algebra-based physics course.* Fall 2025
- *PHY 2053L/2048L, first semester algebra/calculus-based physics labs.* 2023 - 2025

Supplemental Review Session Leader, University of Cincinnati 2021 - 2023
Instructor for MATH 1096, a companion course for calculus 1 and 2, applied calculus.

Math and Science Support Tutor, University of Cincinnati 2021 - 2023
One-on-one tutoring in introductory physics, calculus 1 and 2, and multivariable calculus.

PRESENTATIONS

Talks

- *Vestigial d-wave charge-4e Superconductivity from Bidirectional Pair Density Waves*, APS March Meeting, Mar 2026.
- *Spin fractionalization in a Kondo Lattice Heterostructure*, APS March Meeting, Mar 2023.

Posters

- **E. Huecker**, Y. Wang. *Vestigial d-wave charge-4e Superconductivity from Bidirectional Pair Density Waves*, Quantum Geometry and Moiré Materials Workshop, University of Florida, November 2025

AWARDS AND SCHOLARSHIPS

CCMS Graduate Fellowship for research in condensed matter physics from UF Physics 2024
Physics Scholarship for academic performance from UC Physics 2022
Junior Achievement Award for third year research performance from UC Physics 2022

ADDITIONAL EXPERIENCE

Undergraduate Research Assistant, University of Cincinnati 2022 - 2023

- Developed and analyzed a theoretical model for a Kondo lattice superconductor heterostructure permitting fractionalized spin-excitations. Advised by Dr. Yashar Komijani.
- Investigated the production of scalar leptoquarks from dilepton scattering events using MadGraph collision generation software. Advised by Dr. Conor Henderson.

TECHNICAL SKILLS

Programming Languages MATLAB, Python, C++
Software & Tools Mathematica, LaTeX

SCIENCE OUTREACH

- Reviewer, 2026 Florida STEM Symposium Dec 2025
- Organizer, UF Physics Graduate Student and Postdoc Seminar Series 2025 - Present
- Secretary, UC Society of Physics Students 2022 - 2023

Informal Talks

- *Why Things Move: Forces, Motion, and Gravity*, Scientist in Every Florida School Program, Nov 2026.
- *Quantum Oscillation and Orbital Magnetism of a Flat Band From Quantum Geometry*, UF Physics Graduate Student and Postdoc Seminar, Oct 2026.
- *Vestigial Orders: From Pair Density Waves to Charge-4e Superconductivity*, UF Physics Graduate Student and Postdoc Seminar, Nov 2025.
- *Spin Fractionalization in a Kondo Lattice Superconductor Heterostructure*, UF Physics Graduate Student and Postdoc Seminar, March 2024.
- *Search for T-Channel Production of Scalar Leptoquarks in Dilepton Events*, UC High Energy Physics Group Meeting, Jun 2023.

WORKSHOPS ATTENDED

- Quantum Geometry and Moiré Materials Workshop, *University of Florida.* Nov 2025
- Joint ICTP-WE Heraeus School and Workshop on Advances in Quantum Matter, *Abdus Salam International Centre for Theoretical Physics, Trieste, Italy.* Aug 2025
- Maglab Theory Winter School 2025, *National High Magnetic Field Laboratory, Tallahassee, Florida.* Jan 2025

PROFESSIONAL MEMBERSHIPS

Graduate Member, American Physical Society 2023 - Present